

Environmental Sustainability

Environmental sustainability refers to the responsible use and management of natural resources such as water, air, soil, and energy, in a way that protects ecosystems and ensures that future generations can meet their own needs. It focuses on reducing pollution, conserving biodiversity, minimizing waste, and adopting renewable and clean energy solutions so that human activities do not harm the planet's long-term health.

In the United Arab Emirates, environmental sustainability is a national priority and a core pillar of the country's development vision. Despite its desert climate and limited natural resources, the UAE has taken major steps to protect its environment and promote sustainable growth. Examples include:

- ✓ **Net Zero 2050:** The UAE was the first country in the region to commit to achieving net-zero carbon emissions by 2050.
- ✓ **Clean Energy Projects:** Landmark initiatives such as the Mohammed bin Rashid Solar Park and the Barakah Nuclear Energy Plant support a cleaner energy future.
- ✓ **Water Sustainability:** With scarce freshwater resources, the UAE uses advanced desalination technologies and invests in water recycling and efficient irrigation.
- ✓ **Biodiversity Protection:** The UAE actively protects marine life, desert ecosystems, and endangered species such as the Arabian Oryx.
- ✓ **Sustainable Urban Development:** Smart, green cities such as Masdar City showcase the UAE's commitment to low-carbon, energy-efficient urban planning.
- ✓ **Environmental Awareness Programs:** National strategies such as NEEAS promote sustainability education and community engagement.

Environmental sustainability in the UAE means balancing rapid development with long-term environmental protection, ensuring a healthy, prosperous future for the nation and future generations.

Sustainable Development Goals (SDGs)

The SDGs are a set of 17 global goals established by the United Nations to improve human life and protect the planet by the year 2030. These include:

- No Poverty
- Zero Hunger
- Good Health and Well-being
- Quality Education
- Gender Equality
- Clean Water and Sanitation
- Affordable and Clean Energy
- Decent Work and Economic Growth
- Industry, Innovation, and Infrastructure
- Reduced Inequalities
- Sustainable Cities and Communities
- Responsible Consumption and Production
- Climate Action
- Life Below Water
- Life on Land
- Peace, Justice, and Strong Institutions
- Partnerships for the Goals



1. No Poverty

This goal aims to end all forms of poverty by 2030 through providing fair employment opportunities and ensuring that every individual has an income that allows them to live with dignity. It includes access to adequate housing, healthcare, education, and social protection. Ending poverty is essential for building a stable society capable of facing crises, as poverty is linked to low productivity, higher disease rates, and lower educational attainment.

In the United Arab Emirates, the government has made significant progress in this area. The UAE is considered one of the countries with almost no extreme poverty, thanks to its social welfare system, support for low-income families, and assistance programs run by the Ministry of Community Development, the Zakat Fund, and the Red Crescent. The state also provides affordable housing programs, diverse work opportunities, and initiatives empowering Emiratis economically such as "Nafis" and "Sheikh Zayed Housing Program." Moreover, the UAE supports global efforts to eradicate poverty through international humanitarian aid, ranking among the highest donors relative to its national income.

2. Zero Hunger

This goal focuses on eliminating hunger, ensuring food security, promoting healthy nutrition, and supporting sustainable agriculture. Hunger is not only a lack of food but is also linked to malnutrition, which leads to chronic diseases and reduced productivity. Combating hunger requires strong and resilient food systems capable of withstanding crises.

In the UAE, the country stands as a global model in addressing hunger through heavy investment in food security despite its desert environment. The UAE launched the "National Food Security Strategy," which aims to boost local production through modern technologies such as hydroponics, vertical farming, and smart farms. It also implements the "Ne'ma" initiative to reduce food waste

and supports food banks including the “UAE Food Bank.” Internationally, the UAE leads hunger-relief efforts through its large-scale humanitarian aid to countries in need.

3. Good Health and Well-being

This goal seeks to ensure healthy lives and promote well-being for all age groups. It includes reducing mortality rates, providing high-quality healthcare, combating infectious and non-infectious diseases, and improving mental health. It also involves preventive healthcare and promoting healthy lifestyles.

In the UAE, healthcare is among the best globally. The country has developed an advanced medical system with world-class hospitals, digital health applications, and chronic disease prevention programs. The UAE launched “UAE Vision for Health 2031” to develop innovative preventive and therapeutic services. During the COVID-19 pandemic, the UAE provided one of the fastest and most efficient responses in testing and vaccination. The country also promotes awareness of mental health and encourages healthy lifestyles through initiatives such as “National Sports Day” and “Healthier Community.”

4. Quality Education

This goal aims to ensure inclusive, high-quality education and promote lifelong learning opportunities. Quality education builds a generation capable of innovation and contributing to economic development. It includes improving curricula, training teachers, providing safe learning environments, and integrating technology in education.

In the UAE, education is a core pillar of national development. The government has invested heavily in digital transformation in schools and universities and provided modern, advanced learning environments. Initiatives such as “Madrasa,” which offers thousands of free digital lessons, and the “Mohammed bin Rashid Scholarship Program” support educational excellence. The UAE also enhances higher education and research through globally recognized universities and prioritizes early childhood education and programs for people of determination, making the UAE a regional leader in modern education.

5. Gender Equality

This goal promotes equality between men and women and empowers women and girls in all fields, including education, employment, and political participation. Gender equality enhances fairness, increases productivity, and strengthens society.

In the UAE, women are central partners in development. The country has made global strides in women’s empowerment, with women holding ministerial positions and representing half of the Federal National Council. The UAE provides equitable work opportunities and supports women entrepreneurs through initiatives such as the “Business Women Council.” Laws such as the Family Protection Law reinforce safety and dignity. Emirati women excel academically, forming the majority of university students, and contribute significantly to fields such as technology, medicine, space, and engineering.

6. Clean Water and Sanitation

This goal seeks to ensure access to clean water for all and provide safe and sustainable sanitation services. Water is essential for life, and its shortage leads to serious diseases and affects food production and economic growth.

In the UAE, water is a national priority due to its natural scarcity. The country has built one of the world’s largest desalination networks and works to expand greywater recycling. It continues to modernize sewage systems and uses advanced irrigation technologies to reduce water waste. Awareness initiatives such as “Qatrah” encourage responsible consumption. Internationally, the UAE supports water-related humanitarian projects through organizations like the Red Crescent and “Suqia UAE,” reflecting its strong commitment to sustainability and water security.

7. Affordable and Clean Energy

This goal promotes access to clean, sustainable, and affordable energy for all. Traditional energy sources cause pollution and climate change, prompting the global shift toward renewables such as solar and wind power.

In the UAE, clean energy is a future cornerstone. The country launched major projects like the Mohammed bin Rashid Solar Park, one of the largest in the world, and the “Barakah Nuclear Energy Plant,” which provides carbon-free electricity. Innovation in clean energy is supported by institutions such as “Masdar.” The country also encourages energy efficiency and funds renewable energy projects abroad, contributing to global sustainability.

8. Decent Work and Economic Growth

This goal promotes sustainable economic growth and decent work opportunities while safeguarding workers’ rights and supporting innovation and entrepreneurship.

The UAE has one of the most competitive economies globally. It offers an attractive business environment and strong legal protections for workers. Initiatives such as “Nafis” empower Emiratis in the job market. The UAE is rapidly advancing in the digital economy, innovation, and free economic zones. “UAE Vision 2031” aims to build a diverse, flexible economy focusing on advanced industries like AI, space, and renewable energy.

9. Industry, Innovation, and Infrastructure

This goal focuses on developing advanced industries, improving infrastructure, and promoting scientific research and innovation. The UAE has world-class infrastructure in transportation, airports, seaports, and telecommunications. The country supports industrial development through the “300 Billion Project,” which aims to expand national industries. Innovation is fostered through initiatives like “Dubai Future Accelerators” and the “Ministry of Industry and Advanced Technology.” The UAE’s space achievements, including the “Hope Probe,” reflect its leadership in advanced technologies.

10. Reduced Inequalities

This goal aims to reduce inequalities within and among countries by ensuring fair opportunities, social justice, and support for vulnerable groups.

The UAE is known for its social cohesion and tolerance, hosting more than 200 nationalities living peacefully together. The country provides strong social support programs for citizens and empowers people of determination through policies and inclusive

initiatives. Access to healthcare, education, and fair work opportunities helps minimize inequality. Internationally, the UAE provides humanitarian and developmental assistance to countries in need.

11. Sustainable Cities and Communities

This goal focuses on creating safe, smart, and sustainable cities with modern transport systems, effective waste management, and environmentally conscious urban planning.

In the UAE, urban planning is a national strength. The country has developed smart cities such as Masdar City and “Dubai Smart City.” It offers advanced transport networks including metro systems, trams, electric buses, and e-scooters. The UAE also prioritizes green spaces, air quality improvement, and recycling systems. Building sustainability standards such as “Estidama” in Abu Dhabi and “Al Sa’fat” in Dubai support environmentally friendly construction.

12. Responsible Consumption and Production

This goal encourages sustainable use of resources, waste reduction, and environmentally friendly industrial practices.

In the UAE, numerous initiatives promote responsible consumption, including the “Less Plastic” campaign and Abu Dhabi’s single-use plastic reduction program. The “Ne’ma” initiative combats food waste, while advanced recycling systems support waste management. The UAE also incentivizes industries to adopt clean technologies and reduce emissions, making sustainable production a national priority.

13. Climate Action

This goal stresses the importance of urgent action to combat climate change and reduce carbon emissions. Climate action is a top priority in the UAE. The country announced the Net Zero 2050 strategic initiative, becoming the first in the region to commit to net-zero emissions. The UAE hosted COP28, strengthening global climate collaboration. It also supports renewable energy, electric transport, nuclear power, and sustainable agriculture while promoting environmental awareness across society.

14. Life Below Water

This goal aims to protect oceans and marine life from pollution, overfishing, and environmental degradation.

The UAE has long coastlines along the Arabian Gulf and the Gulf of Oman, and therefore invests heavily in marine conservation. Initiatives include coral reef restoration, turtle protection, beach cleaning campaigns, and reducing marine plastic pollution. The UAE’s “30×30” initiative aims to protect 30% of natural habitats, including marine areas, by 2030. Research centers across the country contribute to marine biodiversity studies and conservation.

15. Life on Land

This goal focuses on protecting terrestrial ecosystems, combating desertification, preserving forests, and protecting biodiversity. Despite its desert environment, the UAE works extensively to protect wildlife. Programs protect Arabian oryx, gazelles, and migratory birds. The UAE has established numerous natural reserves such as the “Dubai Desert Conservation Reserve” and “Al Wathba Reserve.” Large-scale afforestation projects, sustainable agriculture, and advanced water management help combat desertification. Research centers study native plant species to support environmental restoration.

16. Peace, Justice, and Strong Institutions

This goal promotes security, justice, transparency, and effective institutions.

The UAE is one of the safest countries in the world due to its strong, efficient institutions. It provides a fair judicial system, transparent government procedures, and digital services that reduce corruption. The country adopts policies of tolerance and coexistence and even has a dedicated Ministry of Tolerance. The UAE also supports international peace efforts and protects vulnerable communities through humanitarian initiatives.

17. Partnerships for the Goals

This goal highlights the importance of international collaboration to achieve sustainable development.

The UAE embraces global partnerships through cooperation with the United Nations, the World Bank, and developing countries. It hosts major events that promote international cooperation such as Expo 2020 Dubai and Abu Dhabi Sustainability Week. The UAE is one of the world’s largest donors of humanitarian and developmental aid relative to its income, reflecting its commitment to global progress.

The UAE has developed the **National Environmental Education and Awareness Strategy (NEEAS) 2015–2021**, through which it leads initiatives and programs that aim to achieve a healthy and sustainable environment. Six objectives have been set for this national strategy as follows:

1. Educating youth and raising awareness in a way that helps drive the United Arab Emirates towards a sustainable future, and encouraging the integration of the UAE’s environmental priorities into school curricula, in order to create a sense of responsibility for sustainability among young people.
2. Improving society’s commitment to sustainability and environmental protection and creating a sense of collective responsibility, by engaging community ambassadors to raise awareness in different communities through outreach programs.
3. Guiding companies and industries to move towards environmental sustainability, and rewarding and recognizing those companies that fulfill their social responsibility.
4. Engaging key government stakeholders to support environmental sustainability, and ensuring that government entities at all levels serve as role models in adopting sustainable practices.
5. Ensuring the alignment and effectiveness of environmental education and awareness efforts in the United Arab Emirates, by periodically reviewing and tracking the progress made in the strategy while maintaining its consistency with its objectives.
6. Building adequate capacities in the United Arab Emirates to enable the advancement of NEEAS, and establishing effective partnerships between the public and private sectors to implement the vision of the strategy in a productive and efficient manner.

Automation of services and technological development in the UAE

Automation refers to the use of technology, software systems, and intelligent machines to perform tasks with minimal or no human intervention. It aims to improve efficiency, reduce errors, speed up processes, and enhance the overall quality of services across various sectors such as government, healthcare, transportation, education, and industry. Automation relies on tools like sensors, robotics, artificial intelligence, and data-driven systems to ensure that operations run smoothly and consistently, often producing results faster and more accurately than manual methods.

In the United Arab Emirates, automation is a central pillar of the country's digital transformation and smart government initiatives. The UAE has integrated automated systems across many fields, including smart gates at airports, automated toll collection (Salik), smart inventory systems for medical supplies, and AI-powered services in government applications. These innovations support national goals of improving quality of life, increasing government efficiency, and building a knowledge-based economy. By adopting automation at a national scale, the UAE has positioned itself as a global leader in smart services, sustainable development, and future-ready digital infrastructure.

Examples of automation in the United Arab Emirates can be seen across multiple sectors, reflecting the country's commitment to smart transformation and efficient service delivery. Airports use **smart gates** that allow passengers to pass through immigration automatically, while roads rely on the **Salik** automated toll system that deducts fees without stopping vehicles. Government services are heavily automated through applications like the **Smart Employee App**, which enables staff to request leave, approvals, and official documents instantly. In the healthcare sector, Dubai uses automated pharmaceutical inventory systems to prevent medicine shortages and streamline supply management. Transportation services are also automated through apps such as **Abu Dhabi Taxi** and **DTC Smart**, which allow users to request rides using GPS and digital payments. Additionally, UAE cities employ **smart lighting**, **air-quality sensors**, and **smart waste-management systems** as part of their broader smart-city initiatives. These examples highlight how automation has become an integral part of daily life and public services in the UAE.

The UAE Artificial Intelligence Strategy 2031 was launched in 2017 by His Highness Sheikh Mohammed bin Rashid Al Maktoum, Vice President and Prime Minister of the UAE and Ruler of Dubai. This strategy is considered the first major project in the series of initiatives under the UAE Centennial 2071. It consists of five interconnected main pillars (governance, enablement, capacity building, application, and leadership). These pillars represent integrated stages that include development, research, preparation, and the gradual adoption of artificial intelligence technologies and tools across various sectors in the country. This approach will contribute to enhancing performance, accelerating achievement, and creating innovative and creative work environments with high productivity, benefiting from modern technologies. The strategy aims to achieve the following:

- **First Objective:** Establish the UAE as a global destination for artificial intelligence.
- **Second Objective:** Enhance the UAE's competitiveness in priority sectors through the development of AI technologies.
- **Third Objective:** Develop a fertile and supportive ecosystem for artificial intelligence.
- **Fourth Objective:** Adopt AI in customer service sectors to improve quality of life and enhance government performance.
- **Fifth Objective:** Attract and train talent for future jobs that will be enabled by artificial intelligence.
- **Sixth Objective:** Bring world-leading research capabilities to work within targeted sectors.
- **Seventh Objective:** Provide the essential data and supportive infrastructure required to serve as a testing platform for AI.
- **Eighth Objective:** Ensure strong governance and effective regulation.